

Mohamed Deraz Nasr

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Education

Georgia Institute of Technology | 3.9 GPA

Jan 2026 – Dec 2027

M.Sc. Computer Science, Specialization in Computer Perception & Robotics

Coursework: Computer Vision, Deep Learning, Reinforcement Learning, AI for Robotics (localization, SLAM, path planning), Machine Learning

University of Ottawa | 3.7 GPA

Sep 2020 – May 2025

Bachelor of Science (B.Sc.) in Software Engineering

Coursework: Data Structures & Algorithms, OS (threads, processes, memory, synchronization), Embedded/Distributed Systems

Technical Skills

Programming Languages: Python, C++, Rust, Go, Bash, SQL, TypeScript

ML Frameworks: PyTorch, JAX, TensorFlow, TensorRT, ONNX, Hugging Face, NumPy, Pandas, scikit-learn

Systems & Inference: CUDA, OpenVINO, Triton, Ray, Spark, Vulkan, AWS EC2, Docker, Kubernetes, Linux, VTune

Robotics & Simulation: Isaac Lab, Isaac Sim, MuJoCo, Robosuite, Robomimic, COLMAP, ROS2, PyBullet, Open3D

Experience

Qualia

Remote

Research Engineer Intern

Jun 2026 – Present

- Trained Stage-2 residual RLT policy on Pi0.5 (**JAX/PyTorch/Ray/AWS**) for peg insertion in **Isaac Lab**, lifting success **33%** on fixed-ID evals
- Built **Python** residual policy harness over **ZeroMQ/ROS2** to stream actions into **Isaac Lab**, matching **ManiSkill** within **2%** across sim evals

Georgia Institute of Technology

Remote

Graduate Researcher – HAAG Lab

Jul 2026 – Present

- Building 3D mapping pipelines (**VGGT, COLMAP, Neuralangelo, PyTorch, Open3D**) for pose & geometry recovery from multi-view images
- Fusing **point clouds** with SDFs, **3D Gaussian Splatting**, Instant-NGP, and **tiny-cuda-nn** for novel-view synthesis & synthetic training data
- Optimizing **C++/CUDA** GPU memory and latency with **Nsight** while deploying on **Linux** via **Docker, Kubernetes**, and **Git** CI/CD

OpenVINO (Intel) | Google Summer of Code

Remote

ML Systems Engineer

May 2026 – Aug 2026

- Cut Unitree UnifoLM-VLA inference latency **32%** & peak memory **19%** in **C++** (OpenVINO, **OpenCL/oneDNN** GEMM) for live robot control
- Exported Qwen2.5-VL to **ONNX/OpenVINO** at **PyTorch** parity & streamed **ROS2** policies on **Isaac Sim** to **79%** success over **1k+** episodes

Shopify

Ottawa, Canada

Machine Learning Engineer Intern

Jan 2025 – Apr 2025

- Prototyped **PyTorch NLP** text-2-SQL model w/ schema grounding & constrained decoding, reducing querying time by **80%** for **1M+** entries
- Built ML pipelines (**Python/Spark/SQL/Pandas/Numpy/Airflow**) to predict business KPI metrics, improving prediction accuracy by **7%**
- Reduced model response time by **35%** (**Redis, Docker, AWS**) to optimize real time **XGBoost** predictions for merchant dashboards

University of Ottawa

Ottawa, Canada

Undergraduate Research Assistant

Sep 2024 – Dec 2024

- Built **Reinforcement Learning** retraining controller w/ EWMA drift detection & policy evaluation, reducing model recomputation by **40%**
- Engineered **JAX/Python/C++** pipelines processing **200K+** solar readings daily, achieving **80%** data coverage via **Redis** caching
- Developed **React/Go/GraphQL** dashboard serving **XGBoost** forecasts & live model accuracy at **sub-200 ms** latency for live monitoring

Publications

World Model Lens (arXiv 2026) | Unified interpretability for cross-architecture analysis of RSSM, JEPA, & Transformer world models

Affinity Map (Zenodo 2026) | Published meta-learning study for few-shot protein family classification with prototypical networks

Projects

3D Reconstruction Platform – Python, PyTorch, 3D Gaussian Splatting, Nerfstudio, Open3D, COLMAP, cameras, React

- Mapped video to navigable 3D Gaussian scenes: frame extraction, COLMAP camera-pose recovery, and confidence-aware geometry

Motion World Model – Unreal Engine 5, C++, Python, PyTorch, Model Predictive Control, Cross-Entropy Method

- Built real-time model-predictive controller evaluating 64 movement futures, achieving collision free navigation around moving obstacles

Debris Engine – C++, GLSL, OpenGL, Vulkan, GLFW, GLM, Dear ImGui, CMake

- Building robot debris physics sim with spatial-hash collision, contact debug, deterministic replay, and sensor-map export

Reinforcement Learning F1 Sim – Python, PyTorch, Stable-Baselines3, Gymnasium, TensorBoard, Kubernetes

- Built physics-based sim with PPO reinforcement learning & imitation learning, surpassing expert policy by 56% in race reward